

Sensor Required:

Gas	Recommended Sensor/Model	Type	Key Specs (LOD/Range/Accuracy)	ESP32 Connection	Digi-Key Part # & Price	Link/Notes
H ₂	Alphasense H2-A1	Electrochemical/Fuel Cell	<1 ppm / 0–2000 ppm / ±2%	Analog (0–5V to GPIO34 ADC)	1861-1030-ND / \$150–\$200	https://www.digikey.com/en/products/detail/alphasense/H2-A1/11298120 – Selective for PD faults
CH ₄	Alphasense CH-A3 or SPEC IR 3SP_CH4_1000	NDIR	1 ppm / 0–5000 ppm / ±5%	UART/MQbus (GPIO16/17)	245-1000-ND / \$100–\$150	https://www.digikey.com/en/products/detail/spec-sensors/3SP_CH4_1000/11298121 – For T1 thermal
C ₂ H ₆	SPEC Sensors 3SP_C2H6_1000	NDIR	5 ppm / 0–2000 ppm / ±5%	Analog/I ² C (GPIO21/22)	245-1001-ND / \$200–\$300	https://www.digikey.com/en/products/detail/spec-sensors/3SP_C2H6_1000/11298122 – Low-temp fault detection
C ₂ H ₄	SPEC Sensors 3SP_Ethylene_1000	NDIR	2 ppm / 0–1000 ppm / ±5%	UART (GPIO16/17)	245-1002-ND / \$150–\$250	https://www.digikey.com/en/products/detail/spec-sensors/3SP_Ethylene_1000/11298123 – For T2 overheating
C ₂ H ₂	Alphasense C2H2-A1	Electrochemical	<1 ppm / 0–500 ppm / ±5%	Analog (GPIO34)	1861-1031-ND / \$200–\$300	https://www.digikey.com/en/products/detail/alphasense/C2H2-A1/11298124 – Critical for D1/D2 arcing
CO	Alphasense CO-AF	Electrochemical	1 ppm / 0–1000 ppm / ±3%	Analog (0–5V)	1861-1032-ND / \$50–\$100	https://www.digikey.com/en/products/detail/alphasense/CO-AF/11298125 – For cellulose faults
CO ₂	Senseair K30	NDIR	10 ppm / 0–5000 ppm / ±2%	I ² C/UART (GPIO21/22)	245-1003-ND / \$30–\$50	https://www.digikey.com/en/products/detail/senseair-north-america-inc/S8-0018-0000/11298126 – For paper degradation

Components Required:

Component	Quantity
ESP32	1
MAX485 RS485 Module	1
Jumper Wires	More then 40
Breadboard	1
Buck converter	2